

## Diverse Envolving Worlds

### A Standard Simulation Worlds

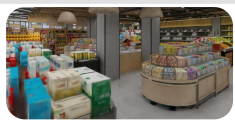


HM3D



MP3D

### B Photorealistic Simulation



Habitat-GS



Interior-GS

### C Evolving Simulation Worlds



HSSD

### D Real-World Evolution



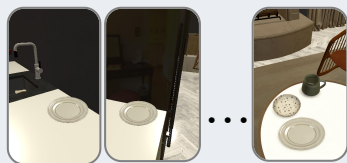
Indoor



Outdoor

Memory  
Construction

#### 1 Observed History



$t_1$

$t_2$

$t_k$

#### 2 Hidden Evolution



world evolves

#### 3 Current Belief



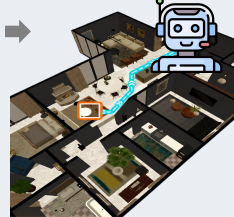
state<sub>1</sub>

state<sub>2</sub>

state<sub>k</sub>

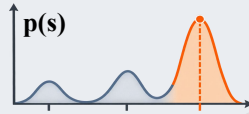
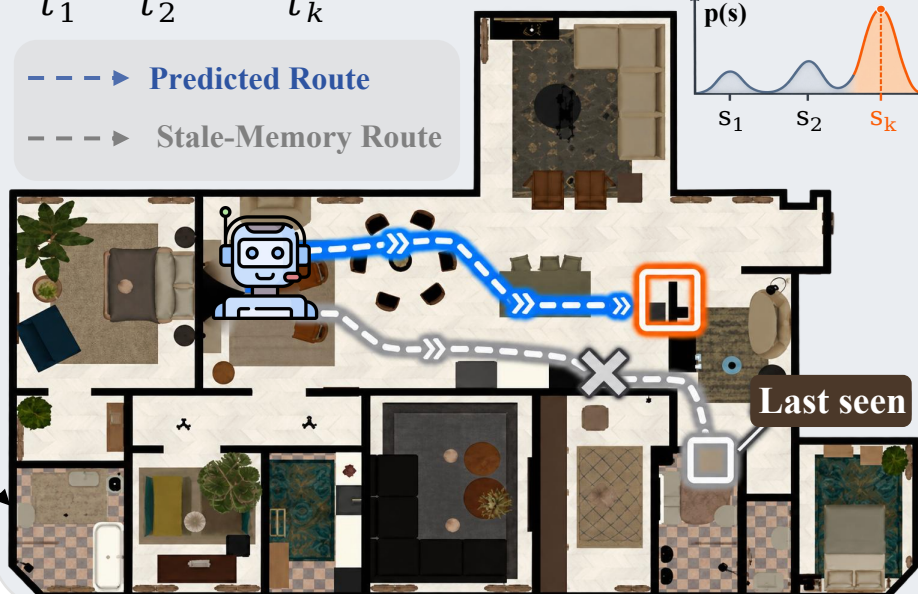
Memory  
Update

#### 4 Belief-guided Plan/Act



---> Predicted Route

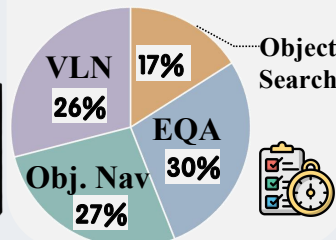
---> Stale-Memory Route



### EvolvingWorld-Bench



54 Scenes  
803.68k Tasks  
4860 Scene-Days  
239.19k Dynamic State Changes



## Unified Embodied Tasks

### A Embodied QA



Q: Where did I put the cup after I left the dining table ?

### B Active Exploration



### C Predictive Navigation



### D Real World Interaction



Task: Find the lounge on this floor and enter it.